A.D. Dulani Sasanka De Silva

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RESEARCH INTERESTS

My research interests are in **natural language processing**, **deep learning, image processing, computer vision** and their applications.

EDUCATION

Bachelor of Science (Honours) in Computer Science Jan 2020 – Jan 2025
University of Peradeniya, Sri Lanka

- **GPA: 3.83/4.00 (First-Class Honours) | Rank (Dept) 3/16 |** Transcript

Wayamba Royal College, Grade 1-13

- G.C.E. Advanced Level Examination: Combined Mathematics, Chemistry, Physics (BBC)
- G.C.E. Ordinary Level Examination: (A-8, B-1)

TEST SCORES

IELTS | Overall Band Score: 7.5

Honours and Awards

University Award for Academic Excellence | PDF

Dean's List- Faculty of Science (2019/2020) Sem I | PDF

Dean's List- Faculty of Science (2020/2021) Sem I | PDF

Dean's List- Faculty of Science (2020/2021) Sem II | PDF

PUBLICATIONS

Temporal aware text driven style transfer for motion-based video transformations| Doc
Proceedings of the Postgraduate Institute of Science Research Congress (RESCON)-2025,
University of Peradeniya, p78 https://ir.lib.pdn.ac.lk/handle/20.500.14444/6013

- Proposed and developed a pipeline integrating object detection, object tracking and segmentation and style transfer to apply text guided styles to objects in a video while preserving the temporal consistency between frames.

CURRENT RESEARCH PROJECTS

Reliable text reading from smart phone captured document videos with vision language models.
Studying how to read text accurately from smartphone videos of documents and books, using vision language models and a self-collected video dataset, with a focus on better tokenization and OCR error analysis.

Privacy preserving code repair with vision language models.
Designing a method to fix non-compiling source code with cloud AI models without exposing the original code, by hiding the code in images that only a fine-tuned vision language model can decode.

TEACHING EXPERIENCE

Temporary Teaching Assistant (Demonstrator) Feb 2025 – Jan 2026

Assisted in conducting practical lab sessions, lectures, assignment evaluations and student support for courses including programming labs (C, Python, Java), Operating system concepts, Object oriented analysis and design, Design and analysis of algorithms, Neural networks and deep learning, Parallel processing, Natural language processing, Software project management.

Visiting Lecturer, NIBM School of Computing March 2025 – Present

Conducted lectures and labs for diploma and undergraduate courses, including Computer Vision, NLP, Database Management Systems and Web Development.

CURRENT APPOINTMENT

Temporary Lecturer Jan 2026 – Present

Statistics and Computer Science Department, University of Peradeniya

INDUSTRIAL EXPERIENCE

Software Engineer, Intern, Focalld Technologies Oct 2024 – Feb 2025

- Developed automation scripts using Selenium (Python) and Playwright (TypeScript) to streamline software testing processes.
- Contributed to the development of an HR Management System using Django (Python).
- Assisted in building a file management system with FastAPI (Python), TypeScript, and Supabase for efficient data handling.

PROJECTS

Quality Analysis of Basmati Rice using Digital Image Processing 2023

Developed an automated image processing system to analyze and grade the quality of Basmati Rice using python, OpenCV.

Opening ceremony of Olympics inside an Olympic Stadium with OpenGL computer graphics 2023

Designed and developed a detailed and realistic architectural simulation of an Olympic Stadium for the opening ceremony of the Olympics with 2D and 3D graphics with C++ and OpenGL.

Management Portal for Toddler Town Preschool and Childcare center 2023

Designed and implemented frontend, backend and Database for Main web page, Inventory system, Payment system of the project using ReactJS, React hooks, NodeJS, MongoDB, JavaScript, CSS, Bootstrap, JSON Web Tokens.

VIP MEET SUITE (COVID free secure meeting room): Computer Device Interfacing (Arduino) 2022

Involved in an innovative design of a secure conference meeting room designed to provide a COVID free environment implemented using microcontrollers, Arduino, sensors, RFID

TECHNICAL STRENGTHS

- **Programming Languages**: Python (Proficient), C, Java, SQL
- **Libraries & frameworks**: TensorFlow, PyTorch, Scikit-Learn, cv2
- **Other**: Git, Linux, Jupyter Notebook, Draw.io, CUDA, OpenMP
- **Soft Skills**: Communication, Teamwork and Collaboration

VOLUNTEERING EXPERIENCE

International Statistical Institute (ISI) - Institute of Applied Statistics Sri Lanka (IASSL) collaborative workshop

- Promotional materials and flyer design 2026 | PDF

Postgraduate Research Congress of Sri Lanka RESCON (PGIS – University of Peradeniya) 2025

- Book editor and Designer – PGIS Research Highlights 2025 | PDF | PDF

Postgraduate Research Congress of Sri Lanka RESCON (PGIS University of Peradeniya) 2026

- Flyer design 2026 | PDF Brochure design| PDF Book Cover design | PDF

CONFERENCES AND COURSES ATTENDED

Conference contribution

- DS&AI 3rd International Conference on Data Science and Artificial Intelligence 19th to 21st November 2025, Kandy Sri Lanka - Participant
- PGIS RESCON 2025 7th and 8th November 2025 - Postgraduate Institute of Science, University of Peradeniya. - Oral Presentation, Participator.

Certificates of completion

- Short course in Human Resource Management - The Open University of Sri Lanka (2023)
- Python course - Centre for Open and Distance Learning (CODL) University of Moratuwa, Sri Lanka (2023)
- Web Design course - Centre for Open and Distance Learning (CODL) University of Moratuwa, Sri Lanka (2023)
- Certificate Course in English for special purposes - English teaching unit, University of Peradeniya. (2023)
- Certificate Course in English for professional purposes - English teaching unit, University of Peradeniya. (2022)
- Course on Career development - Career guidance unit - University of Peradeniya (2021)
- JICA-CHAIR Programme for Japan's Modernization - Sri Lanka, Japan study centre(SLJSC), University of Peradeniya. (2023)

EXTRACURRICULAR ACTIVITIES

Netball Team - College

- Interschool netball championship semi finalist all island A division 2011, 2008
- Interhouse sports meet runners-up (2011) first place (2013)